The Effect of Sport Specialization on Injury Risk in NCAA Athletes: Results From the SAFE Consortium

Jared J. Lutsic, BS,* Seth E. Lutsic, MS,† David S. Ibrahim, DO,* Matthew C. Augusta, DO,‡ Jason D. Robinson, EdD,§ Kevin M. Tong, MS, ATC,|| Mathew R. Saffarian, DO,*¶ Sally E. Nogel, PhD, ATC,** and Nathan J. Fitton, DO*¶

Abstract

Objective: Youth athletes are beginning to specialize in a single sport more often. Previous studies in sports medicine and orthopedics have shown an association between intensity of sport specialization and incidence of injuries. This study is the first of its kind to explore the effects of early sport specialization on injury risk through a multicenter framework with a concentration on NCAA athletics. **Design:** Retrospective cohort study. **Setting:** SAFE Consortium. **Participants:** A total of 211 collegiate athletes from the NCAA's 3 levels of competition: Division I, II, and III. Data were collected by the SAFE investigators. **Intervention:** N/A. **Main Outcome Measures:** Participants completed a questionnaire about their demographics, sport participation, specialization status, physical injuries, recovery period, and treatment method. Specialization status was calculated with a previously published 3-point scale: low, moderate, and high. Injuries were categorized as upper extremity injuries (UEIs) and lower extremity injuries (LEIs). **Results:** Highly specialized athletes were more likely to report UEIs and LEIs than low specialized athletes ($P < 0.0001$). Moderate specialization, in contrast to low specialization, was associated with a higher likelihood of LEIs ($P = 0.03$) but not UEIs ($P = 0.052$). Highly specialized athletes were more likely to report an injury of any kind. **Conclusions:** The SAFE investigators found high specialization was associated with a history of UEIs and LEIs. Return to play was longer for highly specialized athletes versus low specialized athletes (112 days and 85 days, respectively). Highly specialized athletes were more likely to be from Division I and to require surgery.

Key Words: youth athletes, collegiate athletes, sport specialization, physical injuries, clinical practice, training policy

(*Clin J Sport Med* 2024;00:1–5)

INTRODUCTION

Youth sport participation is increasing in North America[1] and Europe.[2,3] Recent studies indicate that more than 44 million youth athletes are participating in some form of organized sports in the United States alone.[4,5] In addition, the amount of NCAA athletes set an all-time record and surpassed 520,000 in the 2021 to 2022 school year.[6] It is demonstrated that there are many advantages to youth sport participation, including physical improvement[7] and psychosocial growth.[8] More precisely, the advantages for youth involve motor skill acquisition,[9] cognitive progress,[10] and psychological development.[11] Hence, abundant advantages exist for youth to participate in organized sports.

Submitted for publication August 21, 2023; accepted April 10, 2024.

*From the *Michigan State University, College of Osteopathic Medicine, East Lansing, Michigan; †Campbell University, School of Osteopathic Medicine, Lillington, North Carolina; ‡University of New England, College of Osteopathic Medicine, Biddeford, Maine; §Lee University, Department of Athletics, Cleveland, Tennessee; ||Defiance College, Department of Kinesiology, Defiance, Ohio; ¶Michigan State University, Department of Orthopedics, East Lansing, Michigan; and **Michigan State University, Department of Athletics, East Lansing, Michigan.*

The authors report no conflicts of interest.

Corresponding Author: Nathan Fitton, DO, Michigan State University, Department of Orthopedics, 4660 South Hagadorn Rd, Suite 420, East Lansing, MI 48823 (fittonna@msu.edu).

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.cjsportmed.com).

Copyright © 2024 Wolters Kluwer Health, Inc. All rights reserved.

http://dx.doi.org/10.1097/JSM.0000000000001225

When youth sport participation turns into intensive, early-age sport specialization, however, there is cause for concern.[12,13] The potential disadvantages span a broad spectrum of topic areas. The common themes include overtraining,[14,15] dropout,[16] burnout,[17] and injuries.[18,19] Among the listed categories, injuries can be further broken down into additional subcategories. For instance, studies have explored the incidence of recurrent,[20] overuse,[21,22] serious overuse,[23] acute,[24] and nonacute[25] injuries. This specific study investigates a handful of the above-referenced subcategories with an emphasis on the incidence of upper extremity injuries (UEIs)[26,27] and lower extremity injuries (LEIs).[28,29] Before concentrating on methods with attention to UEIs and LEIs, it is important to put forth a working definition for sport specialization.

At this point in time, there is not a consistent definition of sport specialization in the literature.[30] For the purpose of this research initiative, sport specialization was defined as "year-round intensive training in a single sport to the exclusion of other sports."[31,32] In short, the above-mentioned definition of sport specialization can be divided into 3 components: (1) single-sport focus, (2) sport exclusion, and (3) year-round involvement. The collegiate athletes were then classified by specialization category based on their responses to a questionnaire. The precise aim of this study was to determine whether an association existed between early sport specialization and the incidence of UEIs and LEIs among collegiate athletes. Data were collected from a large sample of NCAA athletes through the Sport, Action, Finding, and Evaluation (SAFE) Consortium.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

Before undertaking this study, the SAFE investigators hypothesized that highly specialized athletes would have a higher incidence of UEIs and LEIs than those classified under low specialization.

METHODS

The SAFE Consortium is a multisite partnership between an assortment of colleges with the goal to better understand the effects of early sport specialization on athletes' long-term health. For this particular initiative, a retrospective cohort design was selected due to the nature of the data required for principal analysis. This study's protocol was approved by the Institutional Review Board at each site and the Human Research Protection Program, Office of Research Regulatory Support at Michigan State University (HRPP ID: 00007883). Participants provided informed consent before participation. All participants were rostered on a collegiate sports team. The SAFE investigators collected data from all schools in the consortium during the 2022 to 2023 school year. Colleges were chosen based on NCAA division classification (1 Division I, 1 Division II, and 1 Division III) to acquire a diverse sample of collegiate athletes from the NCAA's 3 levels of competition.

Participants

Potential participants were recruited by the SAFE investigators and were provided with study information written by research program staff. All athletes who were rostered in an NCAA sport for the 2022 to 2023 school year were eligible for enrollment in the study. The inclusion criterion for statistical analysis was that each participant had completed the questionnaire in its entirety. Therefore, only incomplete evaluations were excluded from the study.

Questionnaire

All participants completed an anonymous questionnaire that contained a collection of both independent and interconnected items. Participants were asked to report demographic information, sport participation, specialization status, and injury history. Demographic information integrated questions regarding sex, age, high school size, class standing in college, and NCAA division classification. High school size was organized by total enrollment using a published continuum (<500 students = small, 500–1000 students = medium, and >1000 students = large).[33] Average weekly time commitment was calculated in 3 ways. Participants were asked to report the number of hours per week spent in organized sports (team responsibilities), unorganized sports (free-time activities), and any physical education classes (academic events). These questions concentrated on sport participation before and after NCAA registration.

Specialization category was determined by 2 methods. First, participants had to self-identify as either multi-sport or single-sport athletes. Second, research program staff classified participants with an established 3-point scale that has been validated.[34,35] The scale contained the following 3 questions: "Did you pick a main sport?" (single-sport focus), "Did you quit other sports to focus on a main sport?" (sport exclusion), and "Did you train more than 8 months of the year?" (year-round involvement). These questions emphasized sport participation before enrollment onto an NCAA roster. For each of these 3 questions, a "yes" response was counted as 1 point, and a "no" response was counted as 0 points (see Figure 1). Specialization category was then retrospectively assigned according to the total number of points each NCAA athlete accumulated (0-1 = low specialization, 2 = moderate specialization, and 3 = high specialization). Participants were also asked to identify a main sport and record their age at specialization.

Injury history incorporated questions concerning injury rate, injury location (head/neck, shoulder, elbow, hand/wrist, hip/pelvis/thigh, low back, knee, leg, or foot/ankle), injury type (contact, noncontact, overuse, or acute), reinjury rate, surgery rate, and number of missed days of sport participation due to an injury. Research program staff further assigned injury location to UEI and LEI clusters in preparation for statistical analysis. Participants were asked to document any injury before and after registration with the NCAA. While the full questionnaire has not been validated, its components have been used in recent studies.[36,37]

Statistical Analysis

Data were summarized by means and standard deviations (SDs), frequencies and proportions (%), medians and interquartile ranges (IQRs), and *F*-statistics. Models for specialization additionally controlled for hours per week spent in athletic commitments. The primary analysis was a comparison between intensity of sport specialization and likelihood of subsequent physical injury.

Research program staff further used χ^2 tests to determine whether there were associations between groups of 2 categorical variables. ANOVA was used to assess for differences in average weekly time commitment between the 3 specialization categories. Significance was set a priori at $P < 0.05$. Data were analyzed in SPSS statistical software (version 29, IBM Corp).

RESULTS

Of the 240 athletes who started the questionnaire, 211 (92 male, 119 female) completed it in its entirety (average age = 19.98 ± 1.42) (see Figure 2). The majority of the athletes classified themselves as juniors [58 (27.5%)], followed by freshmen [55 (26.0%)], sophomores [50 (23.7%)], seniors [39 (18.5%)], and others [9 (4.3%)] (see Figure 3). Most athletes were from Division II [109 (51.6%)]. The rest were distributed between Division I [66 (31.3%)] and Division III [36

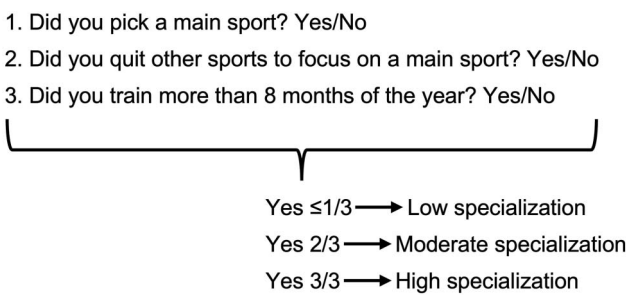

Figure 1. Specialization scale.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

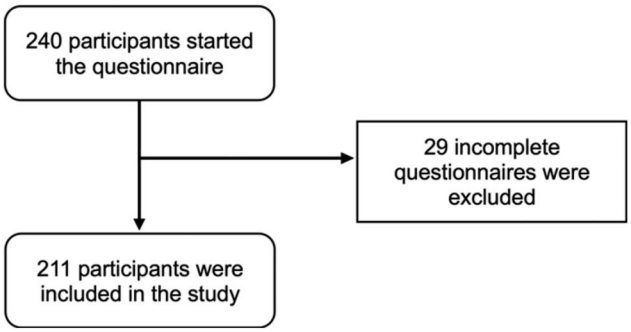

Figure 2. Flow diagram.

(17.1%)]. Specialization status per category was high [102 (48.3%)], moderate [69 (32.7%)], and low [40 (19.0%)] (see Table 1). Notably, athletes from Division I were more likely to be highly specialized than those from Division II [χ^2 (1, N = 175) = 9.37, P = 0.002] and Division III [χ^2 (1, N = 102) = 6.52, P = 0.01]. Average age at specialization was calculated to be slightly above 14 years (14.29 ± 2.21).

Highly specialized athletes were more likely than low specialized athletes to report an injury of any kind [χ^2 (1, N = 142) = 19.81, P < 0.0001]. About three-quarters [157 (74.4%)] of the athletes reported an injury. For athletes who reported an injury, more than half [84 (53.5%)] reported a reinjury. LEIs [108 (61.7%)] were reported more often than UEIs [67 (38.3%)]. High specialization was associated with a greater risk of UEIs [χ^2 (1, N = 142) = 7.17, P = 0.007] and LEIs [χ^2 (1, N = 142) = 4.11, P = 0.04] when compared with low specialized athletes. Moderate specialization was associated with a higher incidence of LEIs [χ^2 (1, N = 109) = 4.69, P = 0.03], but not UEIs [χ^2 (1, N = 109) = 3.78, P = 0.052], in contrast to low specialization. Markedly, there was no significant difference in the number of hours per week spent in athletic commitments between all 3 specializations [F(2, 208) = 0.90, P = 0.41].

TABLE 1. Athlete Characteristics

	N	%
Sex		
Male	92	43.6
Female	119	56.4
NCAA division classification		
Division I	66	31.3
Division II	109	51.6
Division III	36	17.1
Specialization category		
Low	40	19.0
Moderate	69	32.7
High	102	48.3

Over a quarter [57 (27%)] underwent surgery to ameliorate an injury or reinjury. High specialization was associated with a greater likelihood of surgery to treat an injury when compared with low specialization [χ^2 (1, N = 142) = 6.76, P = 0.009]. However, moderate specialization was not more likely to be associated with surgical incidence when evaluated against low specialization [χ^2 (1, N = 109) = 2.32, P = 0.13]. Finally, return to play exhibited a rising trend with each advance in specialization category. For low specialized athletes, they missed 85 days, on average, of practice and competition. Similarly, athletes classified under moderate specialization were away for 110 days. Return to play for highly specialized athletes was the longest at 112 days.

DISCUSSION

To the knowledge of the SAFE investigators, this is the first retrospective cohort study to examine the relationship

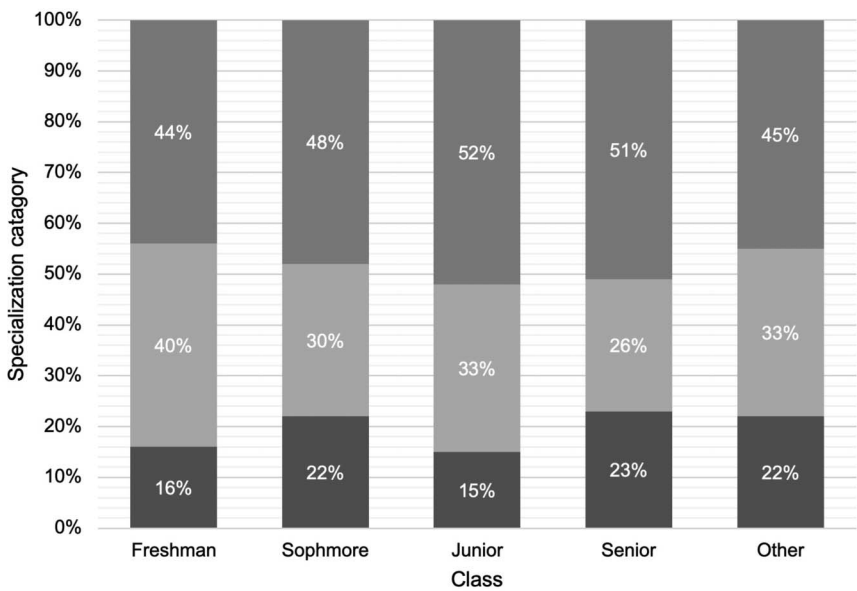

Figure 3. Specialization category by class.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

between intensity of sport specialization and risk of injury through a multicenter framework with a concentration on NCAA athletics. Recent retrospective cohort studies have explored the risk of UEIs and LEIs related to sport specialization; however, some involved only high schools, single colleges, or did not include surgical incidence or return to play.[38,39]

This study adds to the literature both by replicating past results in sports medicine and orthopedics and by providing new information specific to the rate of UEIs and LEIs, surgical incidence, sport participation, and return to play. When contrasted against low specialized athletes, high specialization was associated with a greater likelihood of UEIs and LEIs. Relatedly, highly specialized athletes were more likely to require surgery. Moderate specialization was not associated with a higher incidence of surgery. Furthermore, moderately specialized athletes were more likely to report LEIs but not UEIs. Overall, athletes classified as highly specialized were injured more often than the moderate and low categories. Importantly, there was no significant difference in the number of hours per week spent in athletic commitments across all 3 categories. Finally, return to play increased in an incremental manner from low toward high specialization.

The previously mentioned findings are sufficient to modify existing clinical practice by implementing recommendations at the developmental level of sport. Some specific recommendations relate to end user education, periodization, monitoring, and cross-training. Sports medicine professionals should educate athletes, parents, coaches, and athletic trainers about the impact of high specialization on UEIs, LEIs, and surgical incidence. They should also educate them on the benefits of periodization, which is the structuring of training and competition schedules to allow for peak performance at certain times while emphasizing periods of rest and recovery to prevent overuse injuries.[40]

In addition, athletes should be surveilled closely for signs of overuse injuries by scheduling regular checkups with sports medicine professionals and educating them on the importance of self-monitoring.[41] Finally, athletes should also be strongly encouraged to participate in different sports or activities to engage other muscle groups and prevent overuse; cross-training may also be incorporated into their training regimens. Taken as a whole, these approaches form a robust basis for re-evaluating training practices and developing policies within the SAFE Consortium and beyond.

Limitations

The study has 2 main limitations. First, participants self-reported their own injuries. Second, the questionnaire asked participants about any injuries they sustained. Thus, the data are subject to recall bias. To reduce the potential for recall bias, athletic department personnel and research program staff were available to field any questions from participants as they completed the questionnaire. In an effort to mitigate the limitation of self-reported injuries, future studies could use athletes' medical records to obtain a precise account of their injuries. Furthermore, to address the challenge of recall bias, future studies could be conducted in a longitudinal format where injuries are recorded in real time. This would ensure injuries are reported as they occur rather than relying on memory recall.

Future Directions

The implications of this study for future research are wide-ranging. Since the SAFE investigators examined current athletes retrospectively, only relatively short-term effects of early sport specialization on injuries were ascertained. Further research is recommended to understand the longer-term effects of early-age sport specialization on risk of injury and quality of life. Research is also needed to determine the cross-training activities that most effectively prevent overuse injuries in each sport without diminishing athletic performance. This balance is critical to protecting athletes from injury without decreasing the attainment of athletic scholarships. Finally, since it may be reasonably anticipated that despite all these efforts, some athletes may still specialize early, more research focused on developing and testing interventions to prevent overuse injuries in highly specialized athletes remains necessary.

CONCLUSIONS

In this large sample of NCAA athletes from the SAFE Consortium, highly specialized athletes were more likely to report an injury of any kind. They also missed more days of practice and competition than low specialized athletes. Regarding the types of injuries sustained, highly specialized athletes were at a greater risk of sustaining UEIs and LEIs, despite no significant difference in average weekly time commitment across specialization categories. Among moderately specialized athletes, LEIs, but not UEIs, were more likely to occur when compared with those classified under low specialization. Rates of surgery to treat an injury were significantly greater in highly specialized athletes.

ACKNOWLEDGMENTS

The authors acknowledge the assistance of the Data Manager, Barry R. DeCicco, MS, at Michigan State University, Center for Statistical Training and Consulting. In addition, the authors are grateful for the support of the Director for Research Development, Furqan B. Irfan, MD, PhD, at Michigan State University, College of Osteopathic Medicine.

References

1. Malina RM. Early sport specialization: roots, effectiveness, risks. *Curr Sports Med Rep*. 2010;9:364–371.
2. Mathisen FKS, Kokko S, Tynjala J, et al. Leisure-time physical activity and participation in organized sports: changes from 1985 to 2014 in Finland and Norway. *Scand J Med Sci Sports*. 2019;29:1232–1242.
3. Eithsdottir ST, Kristjansson AL, Sigfusdottir ID, et al. Trends in physical activity and participation in sports clubs among Icelandic adolescents. *Eur J Public Health*. 2008;18:289–293.
4. Memmini AK, Van Pelt KL, Wicklund A, et al. Evaluating adult decision-making modifiers in support of youth contact-sport participation. *J Athl Train*. 2022;57:44–50.
5. Bryan MA, Rowhani-Rahbar A, Comstock RD, et al. Sports- and recreation-related concussions in US youth. *Pediatrics*. 2016;138: e20154635.
6. National Collegiate Athletic Association. NCAA student-athletes surpass 520000, set new record. Available at: https://www.ncaa.org/news/2022/12/5/media-center-ncaa-student-athletes-surpass-520-000-set-new-record.aspx. Accessed March 30, 2023.
7. Brown KA, Patel DR, Darmawan D. Participation in sports in relation to adolescent growth and development. *Transl Pediatr*. 2017;6: 150–159.
8. Mostafavifar AM, Best TM, Myer GD. Early sport specialization, does it lead to long-term problems? *Br J Sports Med*. 2013;47:1060–1061.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

9. Wiersma LD. Risks and benefits of youth sport specialization: perspectives and recommendations. *Pediatr Exerc Sci*. 2000;12:13–22.
10. Waldron S, DeFreese JD, Register-Mihalik J, et al. The costs and benefits of early sport specialization: a critical review of the literature. *Quest*. 2020;72:1–18.
11. Hecimovich M. Sport specialization in youth: a literature review. *J Am Chiro Assn*. 2004;41:32–41.
12. DiFiori JP, Benjamin HJ, Brenner JS, et al. Overuse injuries and burnout in youth sports: a position statement from the American Medical Society for Sports Medicine. *Br J Sports Med*. 2014;48:287–288.
13. Baker J, Cobley S, Fraser-Thomas J. What do we know about early sport specialization? Not much. *High Ability Stud*. 2009;20:77–89.
14. Brenner JS, American Academy of Pediatrics Council on Sports Medicine and Fitness. Overuse injuries, overtraining, and burnout in child and adolescent athletes. *Pediatrics*. 2007;119:1242–1245.
15. Brenner JS. Sports specialization and intensive training in young athletes. *Pediatrics*. 2016;138:154–157.
16. Baker J, Mosher A, Fraser-Thomas J. Is it too early to condemn early sport specialisation? *Br J Sports Med*. 2021;55:179–180.
17. Rugg CM, Coughlan MJ, Li JN, et al. Early sport specialization among former national collegiate athletic association athletes: trends, scholarship attainment, injury, and attrition. *Am J Sports Med*. 2021;49:1049–1058.
18. Normand JM, Wolfe A, Peak K. A review of early sport specialization in relation to the development of a young athlete. *Int J Kin Sports Sci*. 2017; 2:37–42.
19. Post EG, Trigsted SM, Riekena JW, et al. The association of sport specialization and training volume with injury history in youth athletes. *Am J Sports Med*. 2017;45:1405–1412.
20. Confino J, Irvine JN, O'Connor M, et al. Early sports specialization is associated with upper extremity injuries in throwers and fewer games played in Major League Baseball. *Orthop J Sports Med*. 2019;7: 2325967119861101.
21. Myer GD, Jayanthi N, Difiori JP, et al. Sport specialization, part I: does early sports specialization increase negative outcomes and reduce the opportunity for success in young athletes? *Sports Health*. 2015;7: 437–442.
22. Myer GD, Jayanthi N, DiFiori JP, et al. Sports specialization, part II: alternative solutions to early sport specialization in youth athletes. *Sports Health*. 2016;8:65–73.
23. Pasulka J, Jayanthi N, McCann A, et al. Specialization patterns across various youth sports and relationship to injury risk. *Phys Sportsmed*. 2017;45:344–352.
24. Fabricant PD, Lakomkin N, Sugimoto D, et al. Youth sports specialization and musculoskeletal injury: a systematic review of the literature. *Phys Sportsmed*. 2016;44:257–262.
25. Jayanthi NA, Post EG, Laury TC, et al. Health consequences of youth sport specialization. *J Athl Train*. 2019;54:1040–1049.
26. Wilhelm A, Choi C, Deitch J. Early sport specialization: effectiveness and risk of injury in professional baseball players. *Orthop J Sports Med*. 2017; 5:2325967117728922.
27. Post EG, Struminger AH, Hibberd EE. Sport specialization is associated with upper-extremity overuse injury in high school baseball players. *J Shoulder Elbow Surg*. 2020;29:1775–1782.
28. Hall R, Barber Foss K, Hewett TE, et al. Sport specialization's association with an increased risk of developing anterior knee pain in adolescent female athletes. *J Sport Rehabil*. 2015;24:31–35.
29. Ibrahim DS, Lutsic SE, Lutsic JJ, et al. Early sports specialization is associated with an increased risk of lower extremity injuries in NCAA athletes. *PM&R*. 2023;15:S1–S2.
30. Mosher A, Till K, Fraser-Thomas J, et al. Revisiting early sport specialization: what's the problem?. *Sports Health*. 2022;14:13–19.
31. Miller M, Malekian S, Burgess J, et al. Evaluating a commonly used tool for measuring sport specialization in young athletes. *J Athl Train*. 2019; 54:1083–1088.
32. Jayanthi N, Pinkham C, Dugas L, et al. Sports specialization in young athletes: evidence-based recommendations. *Sports Health*. 2013;5:251–257.
33. Post EG, Bell DR, Trigsted SM, et al. Association of competition volume, club sports, and sport specialization with sex and lower extremity injury history in high school athletes. *Sports Health*. 2017;9:518–523.
34. Jayanthi NA, LaBella CR, Fischer D, et al. Sports-specialized intensive training and the risk of injury in young athletes: a clinical case-control study. *Am J Sports Med*. 2015;43:794–801.
35. McGuine TA, Post EG, Hetzel SJ, et al. A prospective study on the effect of sport specialization on lower extremity injury rates in high school athletes. *Am J Sports Med*. 2017;45:2706–2712.
36. Jayanthi N, Kleithermes S, Dugas L, et al. Risk of injuries associated with sport specialization and intense training patterns in young athletes: a longitudinal clinical case-control study. *Orthop J Sports Med*. 2020;8: 2325967120922764.
37. Bell DR, Post EG, Trigsted SM, et al. Prevalence of sport specialization in high school athletics: a 1-year observational study. *Am J Sports Med*. 2016;44:1469–1474.
38. Okoruwa ET, Abbott A, Stamm M, et al. Sport specialization classification and injury risk in female athletes. *Sports Health*. 2022;14:797–804.
39. Ahlquist S, Cash BM, Hame SL. Associations of early sport specialization and high training volume with injury rates in National Collegiate Athletic Association Division I athletes. *Orthop J Sports Med*. 2020;8: 2325967120906825.
40. Mujika I, Halson S, Burke LM, et al. An integrated, multifactorial approach to periodization for optimal performance in individual and team sports. *Int J Sports Physiol Perform*. 2018;13:538–561.
41. Parisien RL, Pontillo M, Farooqi AS, et al. Implementation of an injury prevention program in NCAA Division I athletics reduces injury-related health care costs. *Orthop J Sports Med*. 2021;9:23259671211029898.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.